

SUPERMARKET CHAIN DATA ENGINEERING CAPSTONE PROJECT

SQL & Python Data Pipeline Development

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1. Project Overview and Objectives

This capstone project is designed for 2nd-year Computer Science undergraduate students to develop hands-on data engineering competency using SQL and Python. The project simulates a real-world data engineering scenario where freshers will design and implement a complete data pipeline for a supermarket chain.

1.1 Project Scope

This project encompasses:

- Designing a normalized database schema for a supermarket chain
- Writing efficient SQL queries for data retrieval and analysis
- Building Python ETL (Extract, Transform, Load) pipelines
- Handling data quality issues (duplicates, missing values)
- Creating reports and dashboards for business insights

2. Business Context

2.1 About the Organization

MegaMart Supermarket Chain operates 5 retail stores across Hyderabad, India. Each store serves a different locality with varying customer demographics and product preferences. The stores are located in:

- Store 1 - HITEC City (IT Hub)
- Store 2 - Banjara Hills (Upscale Residential)
- Store 3 - Ameerpet (Commercial Area)
- Store 4 - Kukatpally (Middle-Income Residential)
- Store 5 - Dilsukhnagar (Budget Residential)

2.2 Business Challenges

The organization faces several operational challenges:

- Inventory management across stores - items run out of stock at some locations while overstocked at others
- Sales performance variance - stores in affluent areas show different buying patterns
- Data silos - each store reports data manually without a unified system
- Poor visibility into supplier performance and product profitability
- Inability to track customer purchase patterns and seasonal trends

2.3 Project Goal

Build a centralized data platform that:

- Consolidates sales, inventory, and supplier data from all 5 stores
- Provides insights into sales performance, inventory levels, and supplier reliability
- Enables data-driven decision making for store managers and procurement teams

3. Learning Outcomes

Upon successful completion of this capstone project, students will be able to:

3.1 SQL Competencies

- Design normalized relational database schemas
- Write efficient CRUD (Create, Read, Update, Delete) operations
- Perform complex multi-table JOINS (INNER, LEFT, RIGHT, FULL OUTER)
- Aggregate and group data for analytical queries

3.2 Python Competencies

- Read and parse CSV files containing raw data
- Implement data cleaning and validation logic without external libraries
- Detect and handle data quality issues (duplicates, nulls, outliers)
- Connect to relational databases using Python

3.3 Problem-Solving Competencies

- Analyze business requirements and translate them to technical solutions
- Debug and troubleshoot data pipeline issues
- Optimize queries for performance

4. Dataset Overview

4.1 Data Characteristics

Attribute	Value
Total Records	~10,000 rows
Time Period	Jan 2024 - Dec 2024
Number of Stores	5 locations
Product Categories	12 categories
Database Type	SQLite / PostgreSQL

4.2 Data Tables Description

4.2.1 Stores Table

Contains information about all 5 supermarket store locations.

Column	Data Type	Description	Example
store_id	INT (PK)	Unique store identifier	1-5
store_name	VARCHAR	Name of store	HITEC Store
location	VARCHAR	Area in Hyderabad	HITEC City
manager_name	VARCHAR	Store manager	Rajesh Kumar

4.2.2 Products Table

Contains product catalog information.

Column	Data Type	Description	Example
product_id	INT (PK)	Unique product ID	1001
product_name	VARCHAR	Product name	Basmati Rice 5kg
category	VARCHAR	Product category	Groceries
unit_price	DECIMAL	Price per unit	450.00

4.2.3 Sales Transactions Table

Contains all sales transactions across stores.

Column	Data Type	Description	Example
transaction_id	INT (PK)	Unique transaction	50001
store_id	INT (FK)	Store reference	1
product_id	INT (FK)	Product reference	1001
quantity_sold	INT	Quantity sold	5
sale_date	DATE	Date of sale	2024-01-15

4.2.4 Inventory Table

Tracks stock levels of products in each store.

Column	Data Type	Description	Example
inventory_id	INT (PK)	Unique inventory	1
store_id	INT (FK)	Store reference	1
product_id	INT (FK)	Product reference	1001
quantity_on_hand	INT	Stock level	150

4.2.5 Suppliers Table

Contains supplier information.

Column	Data Type	Description	Example
supplier_id	INT (PK)	Unique supplier	501
supplier_name	VARCHAR	Supplier name	ABC Wholesale
contact_person	VARCHAR	Contact name	Arjun Singh
phone	VARCHAR	Phone number	9876543210

4.2.6 Purchase Orders Table

Tracks purchase orders from suppliers.

Column	Data Type	Description	Example
po_id	INT (PK)	PO identifier	70001
supplier_id	INT (FK)	Supplier ref	501
order_date	DATE	Order date	2024-01-10
delivery_date	DATE	Delivery date	2024-01-15

4.3 Data Quality Issues to Handle

The dataset intentionally includes realistic data quality challenges:

- Duplicate transaction records (same product, store, date, quantity)
- Missing values in manager_name and contact_person fields
- Null entries in delivery_date for cancelled orders
- Incorrect unit prices (negative values, unrealistic amounts)
- Invalid dates (future dates for historical data)
- Store IDs that reference non-existent stores
- Inconsistent data formats (phone numbers with different lengths)

5. Problem Statements

Students will solve 5 interconnected problems with progressive difficulty, building from basic data retrieval to complex analytical insights.

5.1 Problem 1: Data Integration and Schema Design

5.1.1 Problem Statement

MegaMart currently stores data in 5 separate Excel files (one per store) with no centralized database. You have been assigned to design and implement a unified database to consolidate all data.

5.1.2 Tasks

1. Design a normalized relational database schema (3NF - Third Normal Form)
2. Create all necessary tables with appropriate data types and constraints
3. Define primary keys, foreign keys, and relationships
4. Write SQL scripts to create the schema in SQLite/PostgreSQL

5.1.3 Deliverables

- Database schema diagram (ER diagram)
- SQL script file: create_schema.sql
- Documentation explaining design decisions

5.1.4 Success Criteria

- Schema follows normalization principles
- All relationships are properly defined
- Schema can handle data from 5 stores
- Constraints prevent invalid data entry

5.2 Problem 2: Data Loading and Validation

5.2.1 Problem Statement

With the schema ready, you need to load data from CSV files into the database while handling data quality issues. Write Python scripts to extract data from raw files, validate it, clean it, and load it into the database.

5.2.2 Tasks

5. Write Python functions to read CSV files
6. Implement data validation logic (check data types, ranges, formats)
7. Identify and handle duplicates
8. Handle missing values appropriately
9. Implement error handling and logging
10. Load cleaned data into the database

5.2.3 Deliverables

- Python script: data_loader.py

- Python script: data_validation.py
- Log file showing all data quality issues found
- Report documenting data cleaning decisions

5.2.4 Success Criteria

- All 10,000 records successfully loaded
- All duplicates identified and handled
- All missing values addressed
- Validation checks catch invalid data

5.3 Problem 3: Sales Performance Analysis

5.3.1 Problem Statement

Store managers need insights into sales performance. Write SQL queries to analyze sales trends across stores, identify best-selling products, and compare store performance.

5.3.2 Questions to Answer

11. Which store has the highest total revenue in 2024?
12. What are the top 5 best-selling products by quantity?
13. How does each store rank by revenue and transaction count?
14. Which products are most profitable per store?
15. What is the month-on-month sales trend for each store?
16. Which store has the most diverse product sales?

5.3.3 Tasks

17. Write SQL queries to answer each question
18. Use JOINS to combine data from multiple tables
19. Use GROUP BY and aggregate functions (SUM, COUNT, AVG)
20. Create Python script to execute queries and format results
21. Generate summary reports with visualizations

5.3.4 Deliverables

- SQL script: sales_analysis.sql
- Python script: sales_reporter.py
- CSV files with query results
- Summary report with key insights

5.3.5 Success Criteria

- All queries return correct results
- Results are properly aggregated and grouped
- Python script executes without errors
- Insights are actionable for store managers

5.4 Problem 4: Inventory Optimization

5.4.1 Problem Statement

The procurement team needs to optimize inventory levels to minimize stockouts while reducing excess inventory. Analyze current inventory levels, sales velocity, and supplier performance.

5.4.2 Questions to Answer

22. Which products are currently out of stock or low stock (< 50 units)?
23. Which products have the highest sales velocity (fastest moving)?
24. Which products are slow-moving and overstock candidates?

25. What is the average days to deliver for each supplier?
26. Which suppliers are most reliable (on-time delivery rate)?
27. What is the optimal reorder point for each product-store combination?

5.4.3 Tasks

28. Write SQL queries to analyze inventory levels by product and store
29. Calculate sales velocity metrics
30. Analyze supplier delivery performance
31. Implement Python logic to calculate reorder points
32. Generate recommendations for procurement actions

5.4.4 Deliverables

- SQL script: inventory_analysis.sql
- Python script: inventory_optimizer.py
- Reorder recommendations in CSV format
- Supplier performance scorecard

5.4.5 Success Criteria

- Correct identification of stockouts and overstock situations
- Accurate calculation of sales velocity
- Supplier performance metrics are meaningful
- Reorder recommendations are data-driven

5.5 Problem 5: Comparative Store Performance Dashboard

5.5.1 Problem Statement

Senior management needs a comprehensive view of how each store is performing relative to others. Create a data pipeline that computes KPIs for each store and generates a dashboard-ready report.

5.5.2 Metrics to Calculate

33. Total revenue by store
34. Average transaction value
35. Number of unique products sold
36. Inventory turnover ratio
37. Top 3 categories by sales
38. Market share (% of total chain revenue)
39. Year-over-month trend

5.5.3 Tasks

40. Write comprehensive SQL queries for all KPIs
41. Handle edge cases and data quality issues
42. Create Python ETL pipeline to automate calculations
43. Generate multiple output formats (CSV, HTML, PDF)
44. Create comparison visualizations (charts, tables)
45. Document the data pipeline architecture

5.5.4 Deliverables

- SQL script: comprehensive_kpi_queries.sql
- Python script: etl_pipeline.py
- Dashboard data in CSV format
- HTML report with visualizations
- Technical documentation of the pipeline

5.5.5 Success Criteria

- All KPIs calculated correctly
- Pipeline handles data quality issues gracefully
- Output is ready for business users
- Pipeline can be run repeatedly without errors

6. Project Structure and Timeline

6.1 Recommended Project Layout

Create the following directory structure:

- supermarket_project/
 - |— database/ (SQL files)
 - |— data/ (CSV files - raw and processed)
 - |— scripts/ (Python files)
 - |— outputs/ (reports, logs, dashboards)
 - |— documentation/ (technical reports)
 - |— README.md

6.2 Weekly Timeline

Week	Deliverables	Focus Area
1	Schema design, ER diagram, SQL script	Problem 1: Data Integration
1	Data validation, cleaned data, quality report	Problem 2: Data Loading
2	Sales queries, Python reporter, insights	Problem 3: Sales Analysis
2	Inventory analysis, optimizer, recommendations	Problem 4: Inventory
3	KPI queries, ETL pipeline, dashboard HTML	Problem 5: Dashboard
3	Complete documentation, presentation slides	Final: Documentation

7. Evaluation Criteria

7.1 Evaluation Framework

The project will be evaluated on the following criteria:

Criteria	Weight	Excellent	Good
Problem-Solving Approach	35%	Clear decomposition, logical flow	Mostly clear, good approach
SQL Correctness	20%	All queries correct, optimized	Most queries correct
Python Code Quality	20%	Clean code, proper error handling	Working code, minimal errors
Documentation & Reports	15%	Comprehensive, well-organized	Adequate documentation
Presentation	10%	Clear, confident, engaging	Understandable delivery

7.2 Problem-Solving Assessment Focus

The evaluation will particularly focus on:

- How well students break down complex problems into manageable components
- The logical flow and rationale behind their solution approach
- How they handle unexpected data quality issues
- Their ability to debug and troubleshoot issues independently
- How they optimize queries and code for better performance
- Documentation of their reasoning and decision-making process

8. Appendix: Sample Data

8.1 Sample Data - Stores Table

store_id	store_name	location	manager_name
1	HITEC Store	HITEC City	Rajesh Kumar
2	Banjara Store	Banjara Hills	Priya Sharma
3	Ameerpet Store	Ameerpet	Amit Patel
4	Kuka Store	Kukatpally	NULL
5	Dilsuk Store	Dilsukhnagar	Venkat Reddy

8.2 Sample Data - Products Table

product_id	product_name	category	unit_price
1001	Basmati Rice 5kg	Groceries	450.00
1002	Amul Milk 1L	Dairy	55.00
1003	Refined Oil 1L	Cooking Oil	185.00
1004	Turmeric Powder 100g	Spices	45.00

8.3 Data Quality Issues Example

Examples of issues found in the dataset:

- Duplicate: Transaction 50001 appears twice with identical data
- Missing: Manager name is NULL for store_id 4
- Invalid: Product 1045 has unit_price = -100.00
- Inconsistent: Phone numbers in suppliers table have varying formats
- Future Date: Order dated 2025-03-20 (after current year)

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